# Course 02429
# Analysis of correlated data: Mixed Linear Models

## Lecture 11: Repeated measurements I

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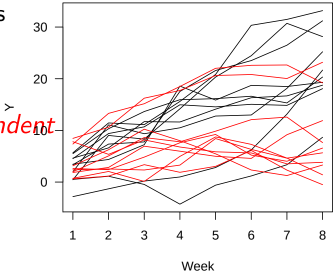
# Outline

# The repeated measurements setup

- Different measurements on the same experimental unit/individual/subject
- Several subjects
- Several measurements on each subject
- Two measurements on the same individual might be correlated
- Might even be highly correlated if "close" and less correlated if "far apart"
- Typical example:
  - 20 individuals from a relevant population
  - Half get drug $A$ and half get drug $B$
  - Measured every week for two months




*To assume that all observations are independent can lead to wrong conclusions/inferences*

# Linear mixed model. No repeated measurements setup.

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \varepsilon, \quad \text{where } \mathbf{u} \sim N(0, \mathbf{G}) \text{ and } \varepsilon \sim N(0, \mathbf{R}),$$
$$E(\mathbf{y}) = \mathbf{X}\beta, \quad \mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R}$$

E.g. Three blocks/groups, two observations per block: $y_{ij} =$

$$\mu + \alpha_i + b_j + \varepsilon_{ij}, \quad b_j \sim N(0, \sigma_B^2), \quad \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2), \quad i = 1, 2, \quad j = 1, 2, 3.$$

$$\mathbf{V} = \begin{pmatrix} \sigma_\varepsilon^2 + \sigma_B^2 & \sigma_B^2 & 0 & 0 & 0 & 0 \\ \sigma_B^2 & \sigma_\varepsilon^2 + \sigma_B^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 + \sigma_B^2 & \sigma_B^2 & 0 & 0 \\ 0 & 0 & \sigma_B^2 & \sigma_\varepsilon^2 + \sigma_B^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_\varepsilon^2 + \sigma_B^2 & \sigma_B^2 \\ 0 & 0 & 0 & 0 & \sigma_B^2 & \sigma_\varepsilon^2 + \sigma_B^2 \end{pmatrix}$$

$$\boldsymbol{ZGZ'} = \begin{pmatrix} \sigma_B^2 & \sigma_B^2 & 0 & 0 & 0 & 0 \\ \sigma_B^2 & \sigma_B^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_B^2 & \sigma_B^2 & 0 & 0 \\ 0 & 0 & \sigma_B^2 & \sigma_B^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_B^2 & \sigma_B^2 \\ 0 & 0 & 0 & 0 & \sigma_B^2 & \sigma_B^2 \end{pmatrix}, \mathbf{R} = \sigma_\varepsilon^2 \mathbf{I} = \begin{pmatrix} \sigma_\varepsilon^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_\varepsilon^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_\varepsilon^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_\varepsilon^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_\varepsilon^2 \end{pmatrix}$$

# Structure in the random effects. Compound symmetry

In LMM we have assumed $\varepsilon \sim N(0, \sigma_\varepsilon^2 I)$

$$\mathbf{R} = \sigma^2 \mathbf{I} = \begin{pmatrix} \sigma_\varepsilon^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_\varepsilon^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_\varepsilon^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_\varepsilon^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_\varepsilon^2 \end{pmatrix}$$

Now, there are more than one observation per subject, we shall assume that these observations are correlated, with a simple correlation structure, compound symmetry, where the correlation between any two measurements in the same subject is the same, e.g. with three measurements per subject:

$$\mathbf{R} = \begin{pmatrix} \sigma_{subj}^2 + \sigma_\varepsilon^2 & \sigma_{subj}^2 & \sigma_{subj}^2 & \cdots & 0 & 0 & 0 \\ \sigma_{subj}^2 & \sigma_{subj}^2 + \sigma_\varepsilon^2 & \sigma_{subj}^2 & \cdots & 0 & 0 & 0 \\ \sigma_{subj}^2 & \sigma_{subj}^2 & \sigma_{subj}^2 + \sigma_\varepsilon^2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma_{subj}^2 + \sigma_\varepsilon^2 & \sigma_{subj}^2 & \sigma_{subj}^2 \\ 0 & 0 & 0 & \cdots & \sigma_{subj}^2 & \sigma_{subj}^2 + \sigma_\varepsilon^2 & \sigma_{subj}^2 \\ 0 & 0 & 0 & \cdots & \sigma_{subj}^2 & \sigma_{subj}^2 & \sigma_{subj}^2 + \sigma_\varepsilon^2 \end{pmatrix}$$

# Example: Activity of rats

Measure the effect of a treatment (drug's concentration) on the activity of two rats in a cage, for ten months. The response measurement was taken once per month ($log(y_{ij}),\ i:1,\ldots,30,\ j:1,\ldots,10$).

| tr | cage | lnc.1 | lnc.2 | lnc.3 | lnc.4 | lnc.5 | lnc.6 | lnc.7 | lnc.8 | lnc.9 | lnc.10 | ln.sum |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 9.93 | 9.64 | 9.76 | 9.60 | 9.32 | 9.25 | 9.07 | 9.21 | 9.17 | 8.83 | 93.79 |
| 1 | 3 | 10.05 | 9.74 | 9.69 | 9.47 | 9.53 | 9.38 | 9.21 | 9.26 | 9.07 | 8.88 | 94.27 |
| 1 | 5 | 9.74 | 9.43 | 9.60 | 9.26 | 9.37 | 9.09 | 9.11 | 9.17 | 8.99 | 9.00 | 92.76 |
| 1 | 7 | 9.87 | 9.87 | 9.93 | 9.72 | 9.69 | 9.50 | 9.26 | 9.27 | 9.16 | 9.16 | 95.42 |
| 1 | 9 | 9.96 | 9.55 | 9.49 | 9.44 | 9.45 | 9.41 | 9.41 | 9.38 | 9.25 | 9.04 | 94.38 |
| 1 | 11 | 10.06 | 9.30 | 9.49 | 9.55 | 9.47 | 9.53 | 9.38 | 9.50 | 9.43 | 9.22 | 94.94 |
| 1 | 13 | 9.50 | 9.38 | 9.44 | 9.62 | 9.57 | 9.52 | 9.21 | 9.35 | 9.39 | 8.97 | 93.95 |
| 1 | 15 | 10.03 | 10.02 | 10.05 | 9.49 | 9.21 | 9.25 | 9.12 | 9.23 | 9.16 | 9.32 | 94.88 |
| 1 | 17 | 9.77 | 9.54 | 9.65 | 9.56 | 9.35 | 9.17 | 9.05 | 9.12 | 9.00 | 8.66 | 92.87 |
| 1 | 19 | 9.83 | 9.44 | 9.64 | 9.22 | 9.13 | 8.90 | 8.81 | 8.88 | 8.89 | 8.65 | 91.39 |
| 2 | 37 | 9.83 | 9.73 | 9.86 | 9.43 | 9.49 | 9.25 | 9.20 | 9.12 | 9.17 | 8.83 | 93.92 |
| 2 | 39 | 9.84 | 9.55 | 9.71 | 9.44 | 9.25 | 9.06 | 9.06 | 9.08 | 8.70 | 8.53 | 92.22 |
| 2 | 41 | 9.62 | 9.53 | 9.72 | 9.58 | 9.54 | 9.29 | 9.20 | 9.13 | 9.16 | 8.62 | 93.38 |
| 2 | 43 | 10.01 | 9.65 | 10.00 | 9.66 | 9.43 | 9.35 | 9.24 | 9.15 | 9.17 | 8.85 | 94.50 |
| 2 | 45 | 9.85 | 9.60 | 9.82 | 9.69 | 9.48 | 9.45 | 9.26 | 9.16 | 9.30 | 9.09 | 94.70 |
| 2 | 47 | 9.52 | 9.23 | 9.50 | 9.26 | 9.13 | 9.09 | 9.15 | 9.16 | 9.05 | 8.64 | 91.73 |
| 2 | 49 | 9.93 | 10.01 | 10.00 | 9.89 | 9.81 | 9.35 | 9.60 | 9.59 | 9.58 | 9.27 | 97.03 |
| 2 | 51 | 9.88 | 9.45 | 9.45 | 8.89 | 9.66 | 9.64 | 9.55 | 9.57 | 9.54 | 9.23 | 94.88 |
| 2 | 53 | 9.35 | 9.00 | 9.78 | 9.73 | 9.76 | 9.73 | 9.62 | 9.51 | 9.57 | 9.07 | 95.12 |
| 2 | 55 | 9.24 | 9.28 | 9.66 | 9.28 | 9.04 | 8.67 | 8.72 | 9.24 | 9.04 | 8.41 | 90.58 |
| 3 | 73 | 9.82 | 9.67 | 9.90 | 9.82 | 10.09 | 9.84 | 9.87 | 10.01 | 9.81 | 9.36 | 98.19 |
| 3 | 75 | 9.69 | 9.37 | 9.71 | 9.83 | 9.61 | 9.60 | 9.58 | 9.56 | 9.47 | 9.11 | 95.53 |
| 3 | 77 | 9.17 | 9.03 | 9.34 | 9.15 | 9.48 | 9.32 | 9.26 | 9.35 | 9.10 | 8.72 | 91.93 |
| 3 | 79 | 9.66 | 9.62 | 9.95 | 9.45 | 9.76 | 9.62 | 9.59 | 9.68 | 9.60 | 9.25 | 96.18 |
| 3 | 81 | 9.67 | 9.49 | 9.95 | 9.94 | 9.89 | 9.86 | 9.85 | 9.81 | 9.62 | 9.38 | 97.45 |
| 3 | 83 | 9.78 | 9.58 | 9.86 | 9.97 | 9.74 | 9.69 | 9.63 | 9.60 | 9.66 | 9.26 | 96.77 |
| 3 | 85 | 9.78 | 8.93 | 9.66 | 9.63 | 9.48 | 9.24 | 9.01 | 9.10 | 9.03 | 8.28 | 92.14 |
| 3 | 87 | 10.46 | 10.28 | 10.49 | 9.62 | 9.19 | 9.22 | 9.11 | 9.37 | 9.07 | 8.50 | 95.31 |
| 3 | 89 | 9.63 | 9.55 | 9.61 | 9.60 | 9.62 | 9.56 | 9.49 | 9.48 | 9.59 | 9.24 | 95.36 |
| 3 | 91 | 9.70 | 9.58 | 9.77 | 9.87 | 10.00 | 9.78 | 9.58 | 9.68 | 9.60 | 9.27 | 96.85 |

# Activity of rats, dataset

```
> str(rats) #long format, one row per measurement
'data.frame': 300 obs. of  5 variables:
 $ treatm: Factor w/ 3 levels "1","2","3": 1 1 1 1 1 1 1 1 1 1 ...
 $ cage  : Factor w/ 30 levels "1","3","5","7",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ month : Factor w/ 10 levels "1","2","3","4",..: 1 2 3 4 5 6 7 8 9 10 ...
 $ lnc   : num  9.93 9.64 9.76 9.6 9.32 ...
 $ monthQ: num  1 2 3 4 5 6 7 8 9 10 ...
```

File in long form:

```
> head(rats) # one row per measurement, 30 x 10 =300
  treatm cage month  lnc monthQ
1      1    1     1 9.93      1
2      1    1     2 9.64      2
3      1    1     3 9.76      3
4      1    1     4 9.60      4
...
```

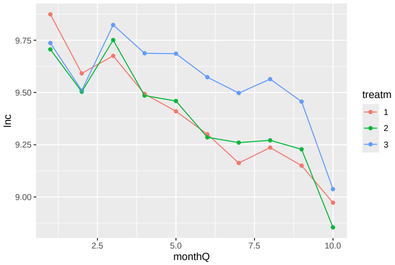
File in short format:

```
> head(rats2) #one row per cage, 30.
  treatm cage lnc.1 lnc.2 lnc.3 lnc.4 lnc.5 lnc.6 lnc.7 lnc.8 lnc.9 lnc.10 ln.sum ln.ave
1      1    1  9.93  9.64  9.76  9.60  9.32  9.25  9.07  9.21  9.17   8.83   93.8   9.38
2      1    3 10.05  9.74  9.69  9.47  9.53  9.38  9.21  9.26  9.07   8.88   94.3   9.43
3      1    5  9.74  9.43  9.60  9.26  9.37  9.09  9.11  9.17  8.99   9.00   92.8   9.28
4      1    7  9.87  9.87  9.93  9.72  9.69  9.50  9.26  9.27  9.16   9.16   95.4   9.54
...
```

# Example: Activity of rats

Summary of the experiment:

- 3 treatments: 1, 2, 3 (concentration)
- 10 cages per treatment
- 10 contiguous months
- The response is activity ($\log($count$)$ of intersections of a light beam during 57 hours)


There are two sources of variability, and the responses are correlated within rats.

# Analysis of the repeated measurements data. Exploratory analysis.

The analysis of the data can be done as:

- Separate analyses for each time-point, 10 analyses.
- One analysis for a summary statistic of the response, one response variable per cage.
- Random effects approach.
- Other statistical methods.

# I. Separate analysis for each time–point. Exploratory analysis.

- Select a fixed time point. The observations at that time (one from each subject) are independent
- Do a separate analysis for the observations at that time
- This can be done for several or for all time–points. However,
  - Difficult to reach a coherent conclusion
  - Sub–tests are not independent
  - Tempting to select time–points that supports out preference
  - Mass significance: If many tests are carried out at 5% level some might be significant by chance. (Bonferroni correction: Use significance level $0.05/n$ instead of $0.05$, $n =$ number of tests.)

# Separate analysis of rats data

- The model at each of the 10 time–points is:

$$\texttt{lnc}_i = \mu + \alpha(\texttt{treatm}_i) + \varepsilon_i \ , \quad \varepsilon_i \sim \text{i.i.d. } N(0, \sigma_\varepsilon^2), \quad i = 1 \ldots 30$$

- The result of the ten tests for no treatment effect:

| **Month** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **F–value** | 1.22 | 0.27 | 1.02 | 2.30 | 3.87 | 4.10 | 4.70 | 7.29 | 4.09 | 0.88 |

Compare with $F_{95\%;2,27} = 3.35$ or $F_{99.5\%;2,27} = 6.49$ if Bonferroni correction is used.

# II. Analysis using a summary statistic. Exploratory analysis.

- Choose a single measure to summarize the individual curves
- This reduces the data set to independent observations
- Popular choices of summary measures:
  - Average over time
  - Slope in regression with time
  - Total increase (last point minus first point)
  - Maximum, minimum point or its difference
- Good method with few and easily checked assumptions
- Good method as an exploratory analysis
- Information may be lost
- Important to choose a good summary measure, specially when the data are unbalanced

# Rats data analyzed via summary measure

- The log of the total activity is chosen as summary measure $\texttt{lnTot} = \log(\text{Total count})$. The data are balanced.

- The one way ANOVA model becomes:

  $$\texttt{lnTot}_i = \mu + \alpha(\texttt{treatm}_i) + \varepsilon_i, \quad \varepsilon_i \sim \text{ i.i.d. } N(0, \sigma_\varepsilon^2), \quad i = 1 \ldots 30$$

- The P–value for no treatment effect in this summary model is 0.0523

- Notice the simplicity of the model and the relative few assumptions

# III. Linear mixed model

- This model uses all observations per subject
- Add subject as a random effect
- Considers measurements on same subject correlated
- Unfortunately equally correlated no matter if they are close or far apart. This can be modified -Next lecture.
- Can be considered as a first step in modelling the actual covariance structure
- Usually only good for short series

# Rats data analyzed via random effects approach

- The model can now be enhanced to:

$$\texttt{lnc}_i = \mu + \alpha(\texttt{treatm}_i) + \beta(\texttt{month}_i) + \gamma(\texttt{treatm}_i, \texttt{month}_i) + d(\mathsf{cage}_i) + \varepsilon_i \ ,$$

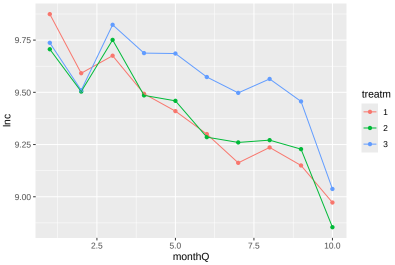
- The covariance structure of this model is:

$$\mathsf{cov}(y_{i_1}, y_{i_2}) = \begin{cases} 0 & , \text{ if } \mathsf{cage}_{i_1} \neq \mathsf{cage}_{i_2} \text{ and } i_1 \neq i_2 \quad (\texttt{different cages}) \\ \sigma_d^2 & , \text{ if } \mathsf{cage}_{i_1} = \mathsf{cage}_{i_2} \text{ and } i_1 \neq i_2 \quad (\texttt{same cage, diff.} \\ \sigma_d^2 + \sigma_\varepsilon^2 & , \text{ if } i_1 = i_2 \quad (\texttt{same measurement}) \end{cases}$$

$\sigma_d^2 = \sigma^2_{subject/cage}$

- The P–value for the interaction term is $0.0059$. Significant, but is the model too simple (random structure)?

# Rats data analyzed via random effects approach (balanced data)


```
m1 <- lmer(lnc~month + treatm + month:treatm + (1 | cage), data = rats)

m2 <- lmer(lnc ~ monthQ + treatm + monthQ:treatm + (1 | cage), data = rats)

m3 <- lmer(lnc ~ monthQ + I(monthQ^2) + treatm + monthQ:treatm + (1 | cage), data = rats)
```

# Rats data analysis, likelihood ratio tests

```
m1 <- lmer(lnc~month + treatm + month:treatm + (1 | cage), data = rats

> ranova(m1)
ANOVA-like table for random-effects: Single term deletions

Model:
lnc ~  month + treatm + (1 | cage) + month:treatm
           npar logLik   AIC   LRT Df Pr(>Chisq)
<none>       32   -4.3  72.6
(1 | cage)   31  -49.4 160.9 90.2  1     <2e-16 ***



> anova(m1)
Type III Analysis of Variance Table with Satterthwaite's method
             Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
month         15.73   1.748     9   243   46.12 <2e-16 ***
treatm         0.24   0.122     2    27    3.22 0.0557 .
month:treatm   1.45   0.080    18   243    2.12 0.0059 **

anova(update(m1, REML=F))
```

# Rats data analysis. Results

```
> summary(m1)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lm
Formula: lnc~-1 + month + treatm + month:treatm + (1 | cage)
   Data: rats

REML criterion at convergence: 8.6

Scaled residuals:
   Min      1Q  Median      3Q     Max
-3.517 -0.454  0.032  0.527  4.093

Random effects:
 Groups   Name        Variance Std.Dev.
 cage     (Intercept) 0.0275   0.166    #Between cages
 Residual             0.0379   0.195     #Within cages
Number of obs: 300, groups:  cage, 30
```

ICC: 0.0275/(0.0275+0.0379)=0.42.

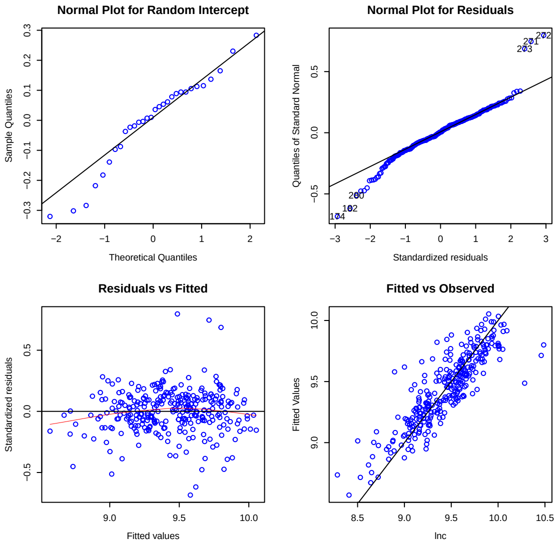
# Rats data analysis, cont.

Alternatively, one can fit the model treating time as a continuous variable: monthQ: 1,..., 10. And include quadratic or higher order terms, e.g.

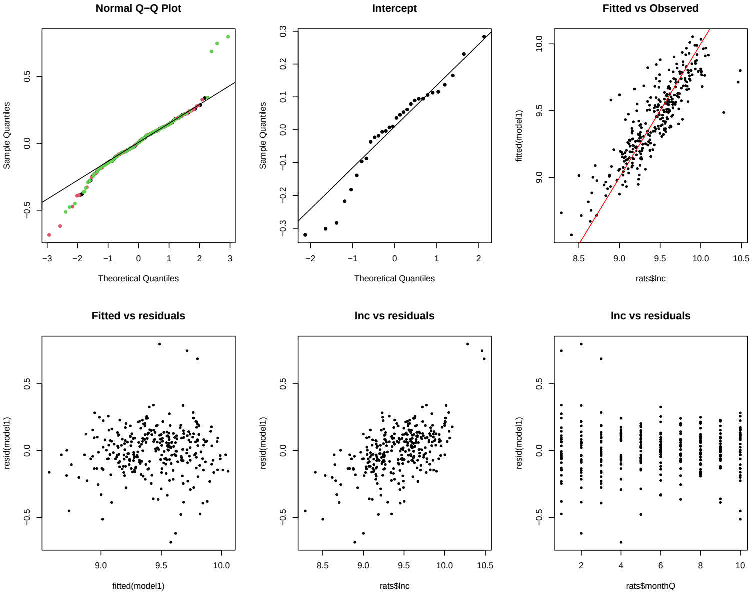
```
 > anova(m1,m2,m3)
refitting model(s) with ML (instead of REML)
Data: rats
Models:
m2: lnc ~ monthQ + treatm + monthQ:treatm + (1 | cage)

m3: lnc ~ monthQ + I(monthQ^2) + treatm + monthQ:treatm + (1 | cage)

m1: lnc ~ month + treatm + month:treatm + (1 | cage)

   npar     AIC    BIC logLik -2*log(L)   Chisq Df Pr(>Chisq)
m2    8   4.854 34.485  5.573   -11.146
m3    9  -1.694 31.640  9.847   -19.694  8.5485  1   0.003458 **
m1   32 -34.789 83.732 49.395   -98.789 79.0949 23  4.447e-08 ***
------
```

Post-hoc analysis with emmeans including plots.

# Rats data analysis. Diagnostic plots

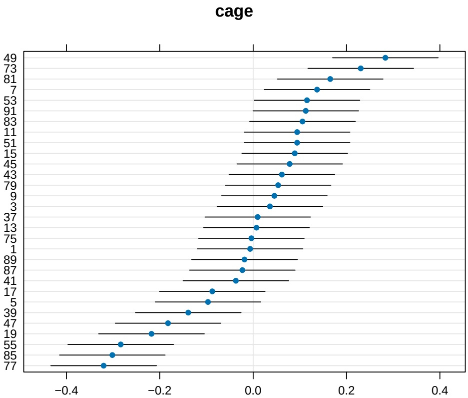


# Rats data analysis. Diagnostic plots

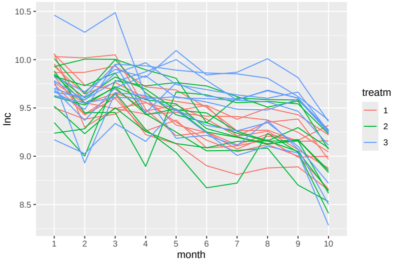


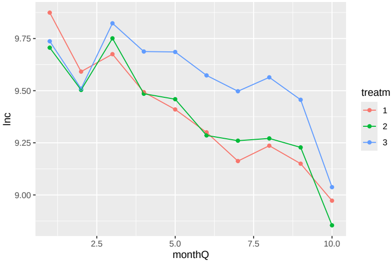
# Rats data analysis

```
> confint(m1)
Computing profile confidence intervals ...
                   2.5 %  97.5 %
.sig01            0.1192  0.2133
.sigma            0.1701  0.2014
...
```



# Rats data analysis. Diagnostic plots




`predict(m1)`
Revise some outlying observations (single measurements).
Exploit/use the model. Post-hoc analysis with `emmeans` including plots.

```
m1 <- lmer(lnc~month + treatm + month:treatm + (1 | cage), data = rats)
```

```
m3: lnc ~ monthQ^2 + I(monthQ^2) + treatm + monthQ:treatm + (1 | cage)
```

The variability across time can not be ignored. The effect of treatment is statistically significant, and the treatment effects vary according to month.

# Fitting a random effects model using libraries lme4 and nlme

I. Fitting a random effects model using function lme4::lmer()/lmerTest::lmer() Covariance with a Compound symmetry structure only.

II. Fit models with different covariance structures using function nlme::lme(). Covariance matrix structure: a) Compound symmetry, b) Gaussian, c) Exponential, d) Linear, others.

III. Fit models with different covariance structures using functionnlme::gls(). Covariance matrix structure: a) Compound symmetry,...

```
m1<- lmer(lnc ~ month + treatm + month:treatm + (1 | cage),
                  REML=F,data = rats)

m1_nlme<- lme(lnc ~ month + treatm + month:treatm, random = ~1 | cage,
                   method = "ML", data = rats)

m1_gls<- gls(lnc ~ month + treatm + month:treatm, correlation=corCompSymm(form=~1|cage),
                  method="ML", data=rats)
```

# Fitting a random effects model using libraries lme4 and nlme

```
m1<- lmer(lnc ~ month + treatm + month:treatm + (1 | cage),
                  REML=F,data = rats)

m1_nlme<- lme(lnc ~ month + treatm + month:treatm, random = ~1 | cage,
                   method = "ML", data = rats)

m1_gls<- gls(lnc ~ month + treatm + month:treatm, correlation=corCompSymm(form=~1|cage),
                  method="ML", data=rats)
```

- The three functions have the option of fitting models with REML and ML.
- gls() fits a linear model using generalized least squares. It allows fitting models with correlated errors and with no random effects. lme() and lmer() do not allow models with no random effects.

# Exercise

Exercise 2 eNote-11.